

1 Question 1

Problem: Consider a graph G consisting of two connected components: C_1 (a complete graph K_{20}) and C_2 (a complete bipartite graph $K_{10,10}$). Calculate the number of triangles in $\bar{G}$, the complement of G .

Solution

Let n be the total number of vertices.

- Component C_1 : K_{20} with $n_1 = 20$ vertices.
- Component C_2 : $K_{10,10}$ with $n_2 = 20$ vertices. Partition sets A and B where $|A| = |B| = 10$.

In the complement graph $\bar{G}$, an edge (u, v) exists if and only if it does not exist in G . The structure of $\bar{G}$ is as follows:

1. **Within C_1 :** Since C_1 is K_{20} in G , it becomes an independent set (no edges) in $\bar{G}$.
2. **Within C_2 :** Since C_2 is $K_{10,10}$ in G , vertices within partition A form a K_{10} in $\bar{G}$, and vertices within partition B form a K_{10} in $\bar{G}$. There are no edges between A and B in $\bar{G}$.
3. **Between C_1 and C_2 :** Since there are no edges between components in G , there are all possible edges between C_1 and C_2 in $\bar{G}$ (forming a $K_{20,20}$).

We count the triangles (sets of 3 mutually connected vertices) in $\bar{G}$ by analyzing possible vertex selections:

Case 1: All 3 vertices in C_1 . There are no edges within C_1 in $\bar{G}$.

$$N_1 = 0$$

Case 2: All 3 vertices in C_2 . Triangles must be entirely within A or entirely within B (since there are no edges between A and B in $\bar{G}$).

$$N_2 = \binom{10}{3} + \binom{10}{3} = 2 \times \frac{10 \cdot 9 \cdot 8}{3 \cdot 2 \cdot 1} = 2 \times 120 = 240$$

Case 3: 2 vertices in C_1 and 1 in C_2 . This requires an edge between the two vertices in C_1 . Since C_1 is an independent set, no such edges exist.

$$N_3 = 0$$

Case 4: 1 vertex in C_1 and 2 in C_2 . Let $u \in C_1$ and $v, w \in C_2$. Edges (u, v) and (u, w) always exist. We form a triangle if edge (v, w) exists. This occurs if v, w are both in A or both in B .

- Choice of u : 20 ways.
- Choice of pair $\{v, w\}$ in A : $\binom{10}{2}$ ways.
- Choice of pair $\{v, w\}$ in B : $\binom{10}{2}$ ways.

$$N_4 = 20 \times \left[\binom{10}{2} + \binom{10}{2} \right] = 20 \times (45 + 45) = 20 \times 90 = 1800$$

Total Triangles:

$$N_{total} = N_2 + N_4 = 240 + 1800 = 2040$$

Answer: There are **2040** triangles in $\bar{G}$.

2 Question 2

Problem: Prove that a non-zero vector $\mathbf{x}$ is a stationary point of the Rayleigh quotient $R(A, \mathbf{x}) = \frac{\mathbf{x}^T A \mathbf{x}}{\mathbf{x}^T \mathbf{x}}$ if and only if $\mathbf{x}$ is an eigenvector of A .

Solution

Let $f(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$ and $g(\mathbf{x}) = \mathbf{x}^T \mathbf{x}$. The Rayleigh quotient is $R(A, \mathbf{x}) = \frac{f(\mathbf{x})}{g(\mathbf{x})}$. We compute the gradient $\nabla R(A, \mathbf{x})$ using the quotient rule:

$$\nabla \left(\frac{f}{g} \right) = \frac{g \nabla f - f \nabla g}{g^2}$$

Given that A is symmetric (adjacency matrix of an undirected graph), the gradients of the numerator and denominator are:

$$\nabla(\mathbf{x}^T A \mathbf{x}) = 2A\mathbf{x}$$

$$\nabla(\mathbf{x}^T \mathbf{x}) = 2\mathbf{x}$$

Substituting these into the quotient rule:

$$\nabla R(A, \mathbf{x}) = \frac{(\mathbf{x}^T \mathbf{x})(2A\mathbf{x}) - (\mathbf{x}^T A \mathbf{x})(2\mathbf{x})}{(\mathbf{x}^T \mathbf{x})^2}$$

A vector $\mathbf{x}$ is a stationary point if $\nabla R(A, \mathbf{x}) = \mathbf{0}$. Since $\mathbf{x} \neq \mathbf{0}$, the denominator is non-zero, so we set the numerator to zero:

$$(\mathbf{x}^T \mathbf{x})A\mathbf{x} - (\mathbf{x}^T A \mathbf{x})\mathbf{x} = \mathbf{0}$$

$$(\mathbf{x}^T \mathbf{x})A\mathbf{x} = (\mathbf{x}^T A \mathbf{x})\mathbf{x}$$

Dividing both sides by the scalar $\mathbf{x}^T \mathbf{x}$:

$$A\mathbf{x} = \frac{\mathbf{x}^T A \mathbf{x}}{\mathbf{x}^T \mathbf{x}} \mathbf{x}$$

$$A\mathbf{x} = R(A, \mathbf{x})\mathbf{x}$$

Let $\lambda = R(A, \mathbf{x})$. The equation becomes:

$$A\mathbf{x} = \lambda \mathbf{x}$$

Conclusion: This is the definition of the eigenvalue equation. Thus, $\nabla R(A, \mathbf{x}) = \mathbf{0}$ if and only if $\mathbf{x}$ is an eigenvector of A , with the corresponding eigenvalue λ being the value of the Rayleigh quotient.

3 Question 3

Problem: Compute the modularity of the two clustering results shown in Figure 1.

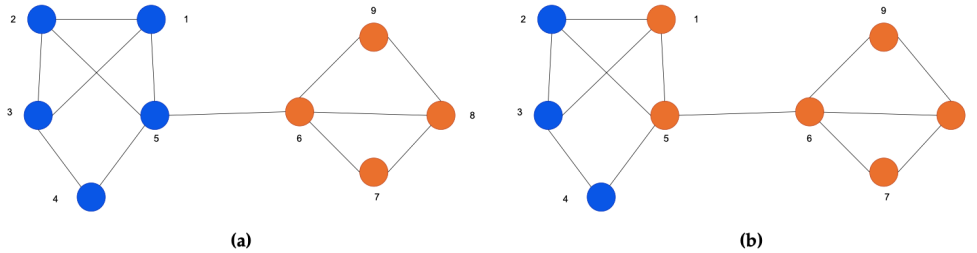


Figure 1: Two graphs where nodes have been assigned to 2 clusters. Cluster membership is indicated by node colour.

Solution

The modularity Q is defined as:

$$Q = \sum_{c=1}^{n_c} \left[\frac{l_c}{m} - \left(\frac{d_c}{2m} \right)^2 \right]$$

Graph Parameters:

- Total edges $m = 13$.
- Total degree $2m = 26$.
- Degrees (k): $k_1 = 3, k_2 = 2, k_3 = 4, k_4 = 2, k_5 = 4$ (Left component), $k_6 = 4, k_7 = 2, k_8 = 3, k_9 = 2$ (Right component).

Partition A (Left Image)

Cluster 1 (Blue): Nodes $\{1, 2, 3, 4, 5\}$

$$l_1 = 7 \quad (\text{Edges within cluster})$$

$$d_1 = 3 + 2 + 4 + 2 + 4 = 15 \quad (\text{Sum of degrees})$$

Cluster 2 (Orange): Nodes $\{6, 7, 8, 9\}$

$$l_2 = 5$$

$$d_2 = 4 + 2 + 3 + 2 = 11$$

Calculation:

$$Q_A = \left[\frac{7}{13} - \left(\frac{15}{26} \right)^2 \right] + \left[\frac{5}{13} - \left(\frac{11}{26} \right)^2 \right]$$

$$Q_A \approx [0.5385 - 0.3328] + [0.3846 - 0.1790]$$

$$Q_A \approx 0.2057 + 0.2056 = \mathbf{0.411}$$

Partition B (Right Image)

Cluster 1 (Blue): Nodes $\{2, 3, 4\}$

$$l_1 = 2$$

$$d_1 = 2 + 4 + 2 = 8$$

Cluster 2 (Orange): Nodes $\{1, 5, 6, 7, 8, 9\}$

$$l_2 = 7$$

$$d_2 = 3 + 4 + 4 + 2 + 3 + 2 = 18$$

Calculation:

$$Q_B = \left[\frac{2}{13} - \left(\frac{8}{26} \right)^2 \right] + \left[\frac{7}{13} - \left(\frac{18}{26} \right)^2 \right]$$

$$Q_B \approx [0.1538 - 0.0947] + [0.5385 - 0.4793]$$

$$Q_B \approx 0.0591 + 0.0592 = \mathbf{0.118}$$

Conclusion:

Partition A ($Q \approx 0.41$) is optimal compared to Partition B ($Q \approx 0.12$).

The significant difference in modularity scores arises from the definition of community structure:

- **Weakest Link Principle:** Partition A cuts the graph at its single "bridge" edge (between nodes 5 and 6). This is the optimal cut because it preserves the two dense cliques, keeping 12 out of 13 edges *internal* to the communities. This yields a high reward in the modularity formula.
- **Breaking Dense Regions:** Partition B forces an artificial boundary through a dense region. By grouping nodes 1 and 5 with the right side, it severs strong internal connections (like 1-2, 1-3, 5-4). This turns many "internal" edges into "inter-community" edges, which modularity heavily penalizes.

4 Question 4

Problem: Give an example of two non-isomorphic graphs mapped to the same representation by the shortest path kernel.

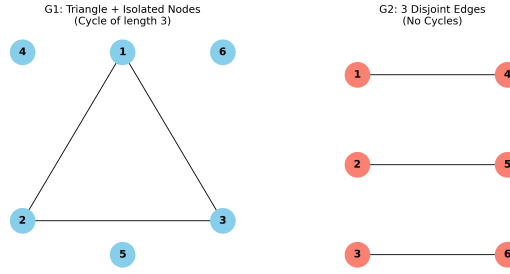


Figure 2: two non-isomorphic graphs same number of nodes considered

Solution

The shortest path kernel counts the number of vertex pairs at shortest-path distance 1, distance 2, etc. To find two non-isomorphic graphs with the same representation, we need two graphs with the exact same histogram of path lengths.

We consider two graphs on 6 vertices:

1. Graph G_1 (Triangle + Isolated Nodes):

- Structure: One triangle (K_3) and 3 isolated vertices.
- Total vertices: 6.

2. Graph G_2 (Perfect Matching / Disjoint Edges):

- Structure: A bipartite graph with partitions of size 3, where every node is connected to exactly one other node (3 disjoint edges).
- Total vertices: 6.

Isomorphism Check: G_1 has a cycle of length 3, whereas G_2 is a bipartite graph with no cycles. Thus, they are non-isomorphic.

Shortest Path Distribution: We calculate the number of pairs (u, v) at specific distances d for both graphs.

For G_1 (Triangle):

- **Distance $d = 1$:** There are 3 edges in the triangle. **Count = 3.**
- **Distance $d = 2$:** In a complete triangle K_3 , every node is connected to every other node. There are no paths of length 2 needed to connect any pair. **Count = 0.**

For G_2 (3 Disjoint Edges):

- **Distance $d = 1$:** There are 3 disjoint edges. **Count = 3.**
- **Distance $d = 2$:** Since the components are just single edges, it is impossible to walk 2 steps. **Count = 0.**

If we consider distance $d = \infty$ (Disconnected): each graph has only 3 connected pairs among all possible pairs $\binom{6}{2}$, meaning $15 - 3 = 12$ disconnected pairs.

Conclusion: Thus both graphs produce the exact same feature vector:

$$\phi = [3, 0, 0, \dots]$$

(3 pairs at distance 1, 0 pairs at any other finite distance). Therefore, the shortest path kernel maps them to the exact same representation.

5 Question 5

Problem: Perform one full iteration of the Weisfeiler–Lehman (WL) subtree kernel algorithm for the labeled graphs G and G' shown in Figure 5. Calculate the kernel value and explain its significance regarding structural similarity.

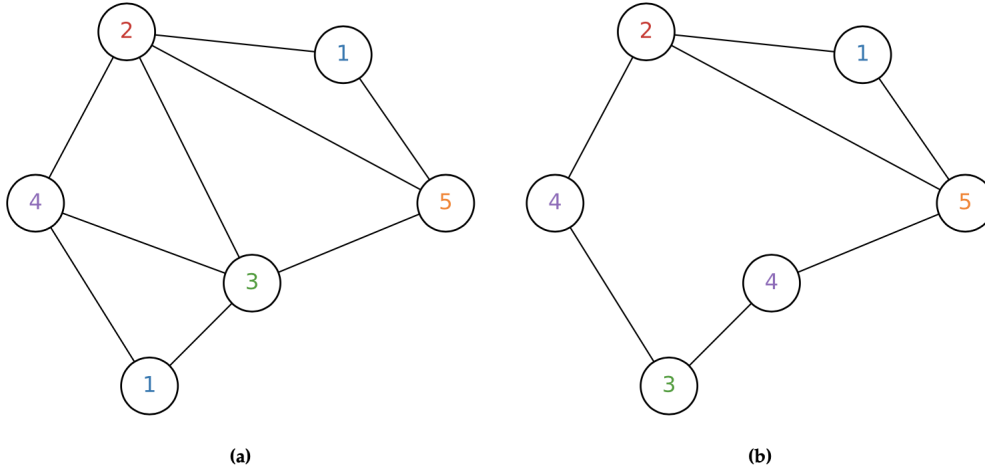


Figure 3: Graphs G and G' , whose nodes are assigned discrete labels.

Solution

The Weisfeiler–Lehman subtree kernel compares graphs by iteratively aggregating neighborhood information. The kernel value after h iterations is defined as:

$$K_{WL}^{(h)}(G, G') = \langle \phi_0(G), \phi_0(G') \rangle + \langle \phi_1(G), \phi_1(G') \rangle + \dots$$

where ϕ_i is the vector of label counts at step i .

1. Graph Analysis

Based on the provided image (colors and numbers), we identify the nodes and edges:

WL Subtree Kernel: One Iteration

Step 0: Initial labels

Each graph has node labels taken from $\{1, 2, 3, 4, 5\}$. The label counts in G and G' are:

Label	G	G'
1	2	1
2	1	1
3	1	1
4	1	2
5	1	1

The 0th-iteration kernel is their dot product:

$$K^{(0)} = \langle \phi_0(G), \phi_0(G') \rangle = 2 \cdot 1 + 1 \cdot 1 + 1 \cdot 1 + 1 \cdot 2 + 1 \cdot 1 = 7.$$

Step 1: WL Iteration (Multisets)

For every node, we create a signature consisting of its own label and the sorted multiset of its neighbors' labels: $S_u = (l_u, \{\text{sorted neighbor labels}\})$.

Signatures for Graph G :

- u_1 (Label 1): Neighbors $\{2, 5\} \rightarrow (1, \{2, 5\}) \rightarrow A$
- u_2 (Label 2): Neighbors $\{1, 3, 4, 5\} \rightarrow (2, \{1, 3, 4, 5\}) \rightarrow B$
- u_3 (Label 5): Neighbors $\{1, 2, 3\} \rightarrow (5, \{1, 2, 3\}) \rightarrow C$
- u_4 (Label 3): Neighbors $\{1, 2, 4, 5\} \rightarrow (3, \{1, 2, 4, 5\}) \rightarrow D$
- u_5 (Label 4): Neighbors $\{1, 2, 3\} \rightarrow (4, \{1, 2, 3\}) \rightarrow E$

- u_6 (Label 1): Neighbors $\{3, 4\} \rightarrow (1, \{3, 4\}) \rightarrow F$

Thus G has the new labels:

$$A, B, C, D, E, F$$

each appearing exactly once.

Signatures for Graph G' :

- v_1 (Label 1): Neighbors $\{2, 5\} \rightarrow (1, \{2, 5\}) \rightarrow A$
- v_2 (Label 2): Neighbors $\{1, 4\} \rightarrow (2, \{1, 4\}) \rightarrow G$
- v_3 (Label 5): Neighbors $\{1, 4\} \rightarrow (5, \{1, 4\}) \rightarrow H$
- v_4 (Label 4): Neighbors $\{3, 5\} \rightarrow (4, \{3, 5\}) \rightarrow I$
- v_5 (Label 3): Neighbors $\{4, 4\} \rightarrow (3, \{4, 4\}) \rightarrow J$
- v_6 (Label 4): Neighbors $\{2, 3\} \rightarrow (4, \{2, 3\}) \rightarrow K$

Thus G' has:

$$A, G, H, I, J, K.$$

Step 2: Compute kernel:

The only new WL label that appears in *both* graphs is $A = (1, \{2, 5\})$:

The 1st-iteration kernel is the dot product:

$$K^{(1)} = 1 \cdot 1 = 1.$$

Total WL kernel after one iteration:

$$K(G, G') = K^{(0)} + K^{(1)} = 7 + 1 = 8.$$

Interpretation of Structural Similarity

The kernel results show that although G and G' share very similar initial node-label distributions, yielding a relatively high step-0 score of 7, their structural similarity diminishes sharply once local neighborhoods are taken into account. After one WL iteration, only a single refined neighborhood pattern matches between the two graphs, resulting in a very low step-1 score of 1. This means that the graphs contain the same labels but two different topologies: G exhibits a denser, more interconnected structure, whereas G' has a much simpler and more linear arrangement. Thus, the WL kernel can successfully detect, despite graphs having similar nodes, significant structural organization differences after just one iteration.